

ASSIGNMENT

- 1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?**

Native development creates separate apps for Android and iOS, offering the best performance and full access to device features, making it ideal for high-performance apps like games and banking applications. Hybrid development uses web technologies in a single codebase, making it faster and more affordable for simple apps and prototypes. Cross-platform development uses frameworks like Flutter to build one codebase for multiple platforms, providing a balance of performance, cost, and development speed, making it suitable for most modern business and mobile applications.

- 2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?**

The Flutter ecosystem is a collection of tools, libraries, widgets, and plugins that help developers create applications for Android, iOS, web, and desktop using a single codebase. It makes development faster, provides a consistent look and feel across different platforms, and delivers performance that is close to native apps, making it a practical and efficient choice for building modern mobile applications.

- 3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?**

The Software Development Life Cycle (SDLC) is a step-by-step process used to plan, design, develop, test, deploy, and maintain a software application. It provides developers with a clear guide throughout the entire project, ensuring that each phase is completed properly before moving to the next. By following the SDLC, development teams can identify and fix problems early, reduce risks and costs, meet user requirements, and produce high-quality, reliable software. This structured approach also improves teamwork, project management, and the overall success of the application.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Flutter development practices and the Software Development Life Cycle (SDLC) work together to make the app development process more organized and efficient. The SDLC provides a clear step-by-step process—from planning and designing the application to developing, testing, deploying, and maintaining it—while Flutter allows developers to build applications for Android and iOS using a single codebase. By combining Flutter with the structured approach of the SDLC, developers can create applications more quickly, reduce errors, save time and development costs, and ensure the final product is reliable, scalable, and meets the needs of its users.

5. Why should developers understand these concepts before starting an application project?

Developers should understand these concepts before starting an application project because they help them choose the most suitable development approach, use the right tools and technologies, and follow a clear and organized development process. Knowing the differences between native, hybrid, and cross-platform development, understanding the Flutter ecosystem, and applying the Software Development Life Cycle (SDLC) allow developers to make better decisions throughout the project. This helps reduce errors, save time and development costs, improve teamwork, and ensure the application is reliable, efficient, and meets the needs and expectations of its users.